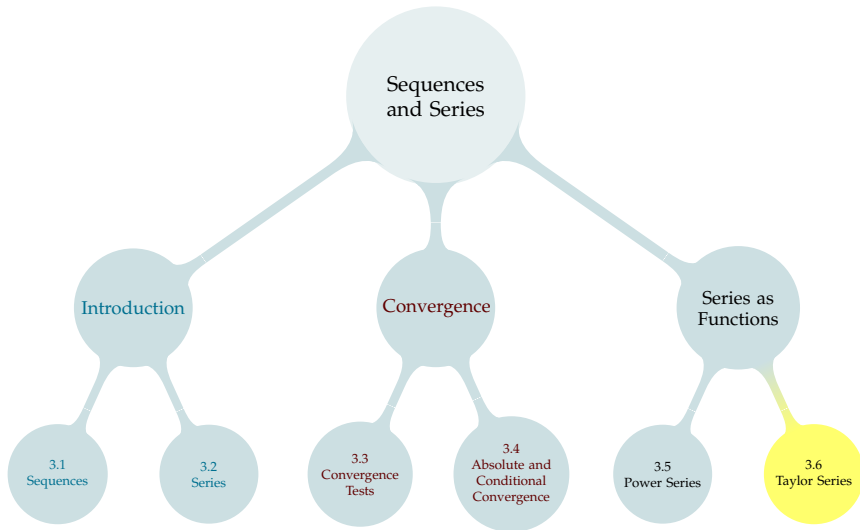


# TABLE OF CONTENTS



## Taylor polynomial

Let  $a$  be a constant and let  $n$  be a non-negative integer. The  $n^{\text{th}}$  order Taylor polynomial for  $f(x)$  about  $x = a$  is

$$T_n(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(a) \cdot (x - a)^k.$$

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The Taylor series for the function  $f(x)$  expanded around  $a$  is the power series

$$\sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(a) (x - a)^n.$$

When  $a = 0$  it is also called the Maclaurin series of  $f(x)$ .

Let's compute some Taylor series, using the definition.

The method is nearly identical to finding Taylor *polynomials*, which is covered in CLP-1.

Find the Maclaurin series for  $f(x) = \sin x$ .

Find the Maclaurin series for  $f(x) = \cos x$ .

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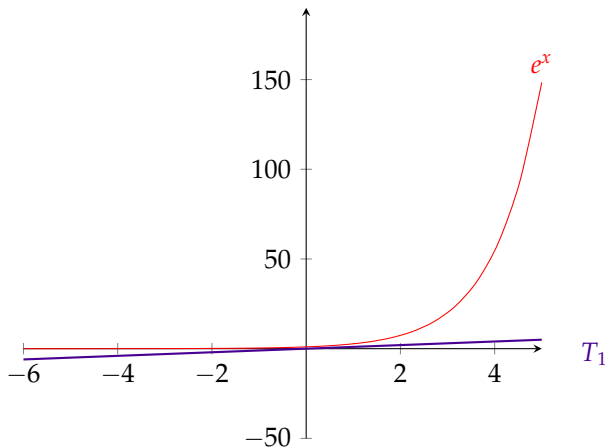
The Maclaurin series for  $f(x) = e^x$  is:

Let  $T_n(x)$  be the  $n$ -th order Taylor polynomial of the function  $f(x)$ , centred at  $a$ .

When we introduced Taylor polynomials in CLP-1, we framed  $T_n(x)$  as an approximation of  $f(x)$ .

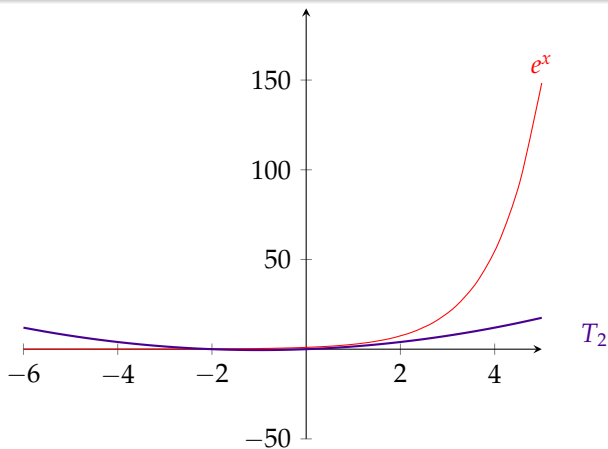
Let's see how those approximations look in two cases:

# TAYLOR POLYNOMIALS FOR $e^x$



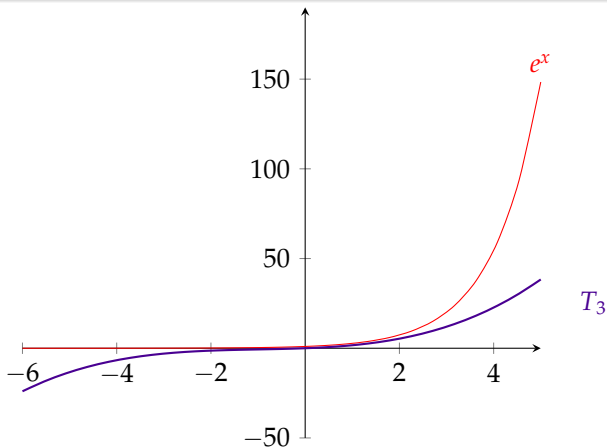
$$e^x \approx 1 + x \quad \text{for } x \approx 0 \quad (\text{linear approximation})$$

# TAYLOR POLYNOMIALS FOR $e^x$



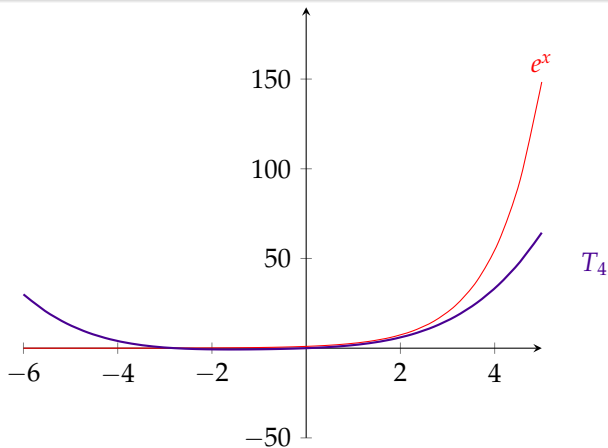
$$e^x \approx 1 + x + \frac{x^2}{2} \quad \text{for } x \approx 0 \quad (\text{quadratic approximation})$$

# TAYLOR POLYNOMIALS FOR $e^x$



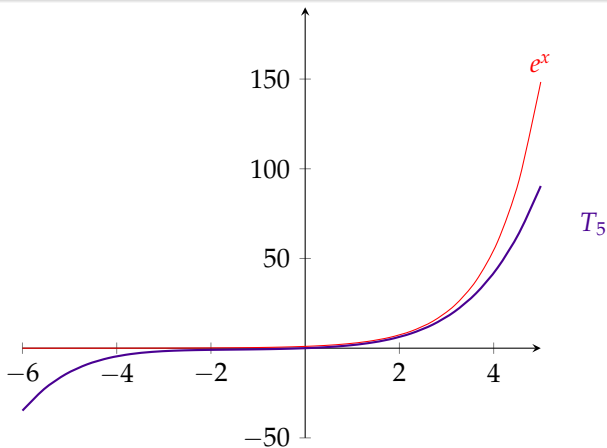
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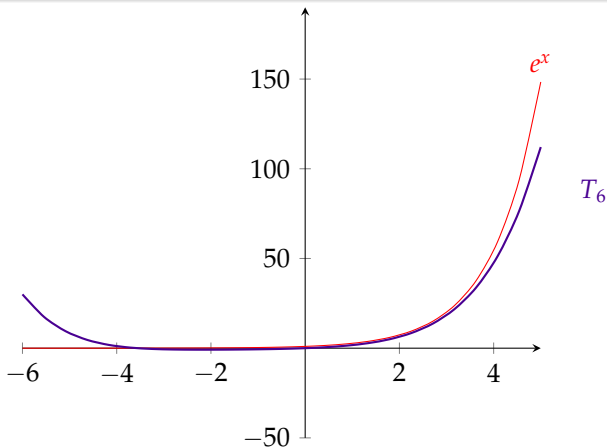
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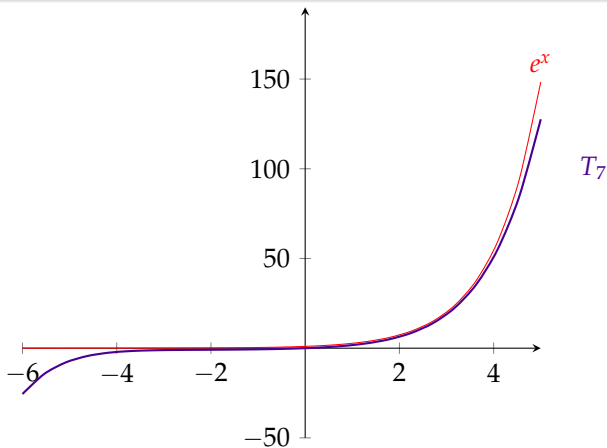
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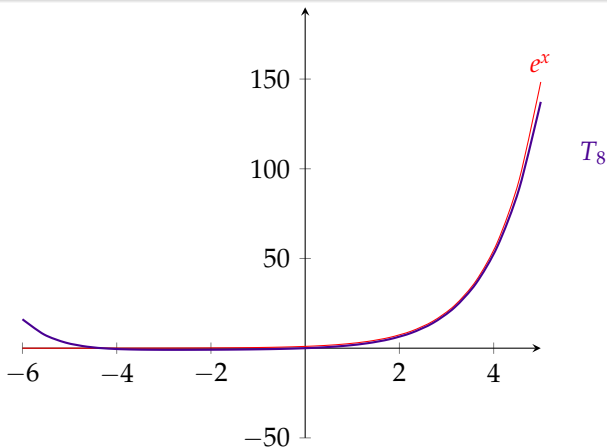
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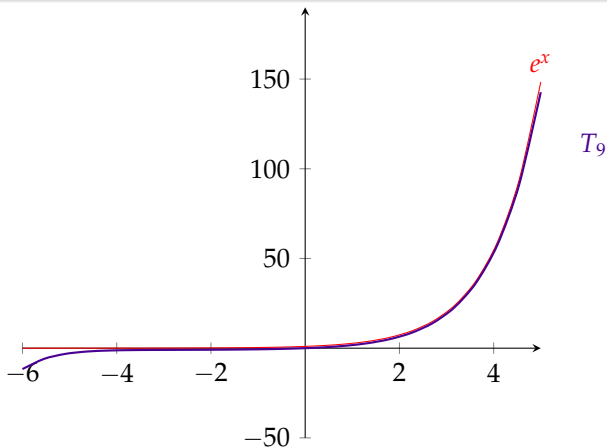
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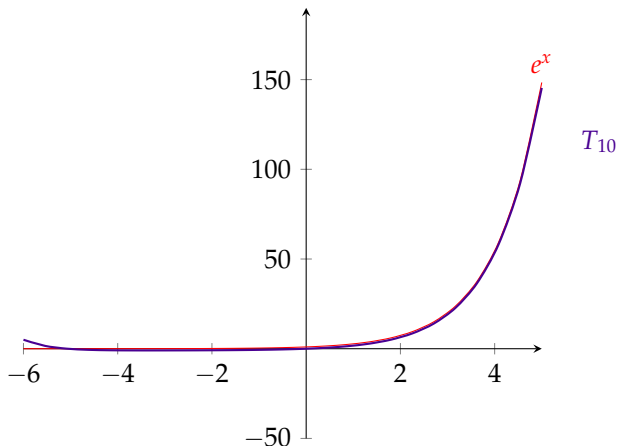
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$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \frac{x^8}{8!} + \frac{x^9}{9!} \quad \text{for } x \approx 0$$

# TAYLOR POLYNOMIALS FOR $e^x$



It seems like high-order Taylor polynomials do a pretty good job of approximating the function  $e^x$ , at least when  $x$  is near enough to 0.

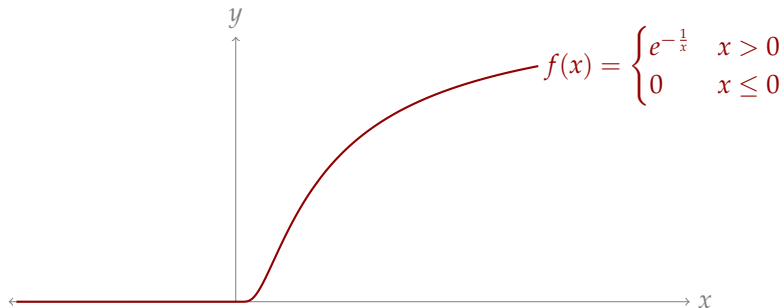
# TAYLOR POLYNOMIALS FOR A DIFFERENT FUNCTION

But that is not the case for all functions. Define

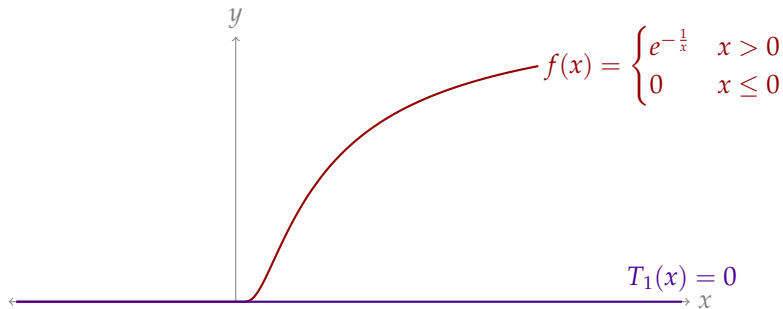
$$f(x) = \begin{cases} e^{-\frac{1}{x}} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Using the definition of the derivative and l'Hôpital's rule, one can show that  $f^{(n)}(0) = 0$  for all natural numbers  $n$ .

# TAYLOR POLYNOMIALS FOR A DIFFERENT FUNCTION

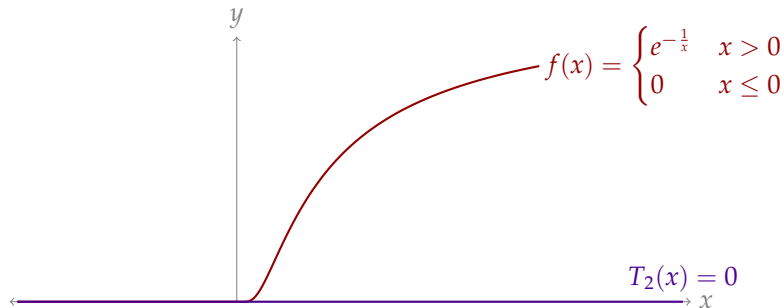


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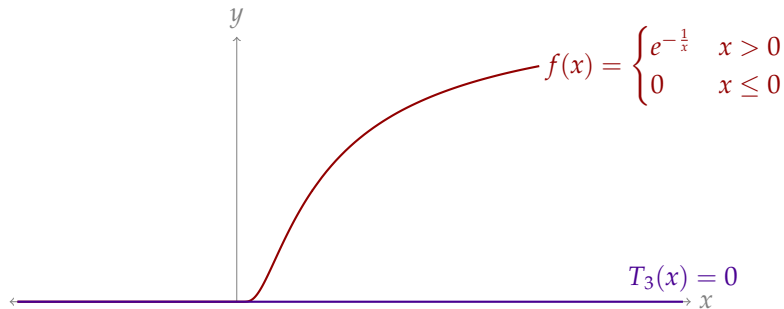
$$f(x) \approx 0 + 0x = 0 \quad \text{for } x \approx 0 \quad (\text{linear approximation})$$

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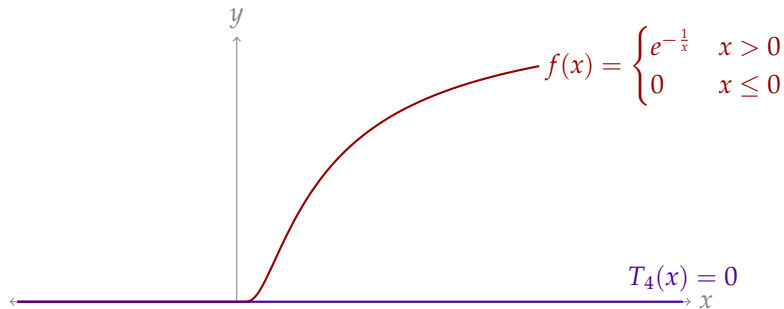
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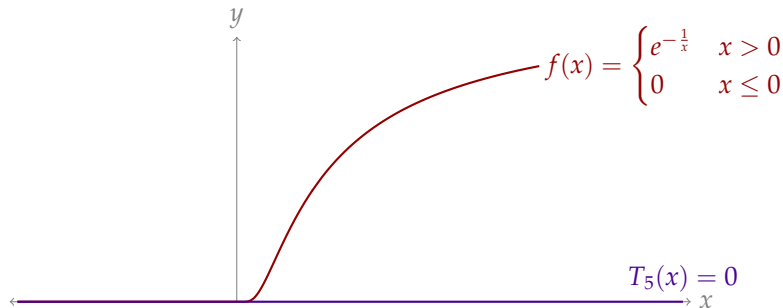
$$f(x) \approx 0 + 0x + 0\frac{x^2}{2} + 0\frac{x^3}{3!} = 0 \quad \text{for } x \approx 0 \quad (\text{cubic approximation})$$

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$$f(x) \approx 0 + 0x + 0\frac{x^2}{2} + 0\frac{x^3}{3!} + 0\frac{x^4}{4!} = 0 \quad \text{for } x \approx 0 \quad (\text{quartic approximation})$$

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$$f(x) \approx 0 + 0x + 0\frac{x^2}{2} + 0\frac{x^3}{3!} + 0\frac{x^4}{4!} + 0\frac{x^5}{5!} = 0 \quad \text{for } x \approx 0 \quad (\text{quintic approximation})$$

Taylor polynomial approximations don't **always** get better as their orders increase – it depends on the function being approximated.

# INVESTIGATION

► We found the Maclaurin series for  $f(x) = e^x$  is  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ .

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- ▶ But, it's not immediately clear whether  $e^x \stackrel{?}{=} \sum_{n=0}^{\infty} \frac{x^n}{n!}$ .
- ▶ We're going to demonstrate that  $e^x$  is in fact equal to  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ . The proof involves a particular limit:  $\lim_{n \rightarrow \infty} \frac{|x|^n}{n!}$ . We'll talk about that limit first, so that it doesn't distract us later.

Intermediate result:  $\lim_{n \rightarrow \infty} \frac{|x|^n}{n!}$ , when  $x$  is some fixed number.

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For large  $n$ , we can think of  $\frac{|x|^n}{n!}$  as a long multiplication, with decreasing terms. At some point, those terms are all decreasing *and less than 1*.

$$\begin{aligned} \frac{|x|^n}{n!} &= \frac{|x| \cdot |x| \cdot |x| \cdot |x| \cdot |x| \cdot |x| \cdot \dots \cdot |x|}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot n} \\ &= \left(\frac{|x|}{1}\right) \left(\frac{|x|}{2}\right) \left(\frac{|x|}{3}\right) \left(\frac{|x|}{4}\right) \left(\frac{|x|}{5}\right) \left(\frac{|x|}{6}\right) \dots \left(\frac{|x|}{n}\right) \end{aligned}$$

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$$\begin{aligned} \frac{|2|^n}{n!} &= \frac{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot \dots \cdot 2}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot n} \\ &= \left(\frac{2}{1}\right) \left(\frac{2}{2}\right) \left(\frac{2}{3}\right) \left(\frac{2}{4}\right) \left(\frac{2}{5}\right) \left(\frac{2}{6}\right) \cdots \left(\frac{2}{n}\right) \end{aligned}$$

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$$\begin{aligned} \frac{|3|^n}{n!} &= \frac{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot \dots \cdot 3}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot n} \\ &= \left(\frac{3}{1}\right) \left(\frac{3}{2}\right) \left(\frac{3}{3}\right) \left(\frac{3}{4}\right) \left(\frac{3}{5}\right) \left(\frac{3}{6}\right) \cdots \left(\frac{3}{n}\right) \end{aligned}$$

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$$\begin{aligned} \frac{|-4|^n}{n!} &= \frac{4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot \dots \cdot 4}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot n} \\ &= \left(\frac{4}{1}\right) \left(\frac{4}{2}\right) \left(\frac{4}{3}\right) \left(\frac{4}{4}\right) \left(\frac{4}{5}\right) \left(\frac{4}{6}\right) \cdots \left(\frac{4}{n}\right) \end{aligned}$$

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$$\begin{aligned} \frac{|\pi|^n}{n!} &= \frac{\pi \cdot \pi \cdot \pi \cdot \pi \cdot \pi \cdot \pi \cdot \dots \cdot \pi}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot n} \\ &= \left(\frac{\pi}{1}\right) \left(\frac{\pi}{2}\right) \left(\frac{\pi}{3}\right) \left(\frac{\pi}{4}\right) \left(\frac{\pi}{5}\right) \left(\frac{\pi}{6}\right) \cdots \left(\frac{\pi}{n}\right) \end{aligned}$$

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$$\begin{aligned} \frac{|x|^n}{n!} &= \frac{|x| \cdot |x| \cdot |x| \cdot |x| \cdot |x| \cdot |x| \cdot \dots \cdot |x|}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \dots \cdot n} \\ &= \left(\frac{|x|}{1}\right) \left(\frac{|x|}{2}\right) \left(\frac{|x|}{3}\right) \left(\frac{|x|}{4}\right) \left(\frac{|x|}{5}\right) \left(\frac{|x|}{6}\right) \dots \left(\frac{|x|}{n}\right) \end{aligned}$$

We're multiplying terms that are closer and closer to 0, so it seems quite reasonable that this sequence should converge to 0.

For a more formal proof, we can use the squeeze theorem to compare this sequence to a geometric sequence.

Intermediate result:  $\lim_{n \rightarrow \infty} \frac{|x|^n}{n!}$ , when  $x$  is some fixed number.

Let  $\frac{|x|}{k}$  be the first factor that's less than 1. Then when  $n > k$ :

# INVESTIGATION

- ▶ We found the Maclaurin series for  $f(x) = e^x$  is  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ .
- ▶ But, it's not immediately clear whether  $e^x \stackrel{?}{=} \sum_{n=0}^{\infty} \frac{x^n}{n!}$ .  
How could we determine this?

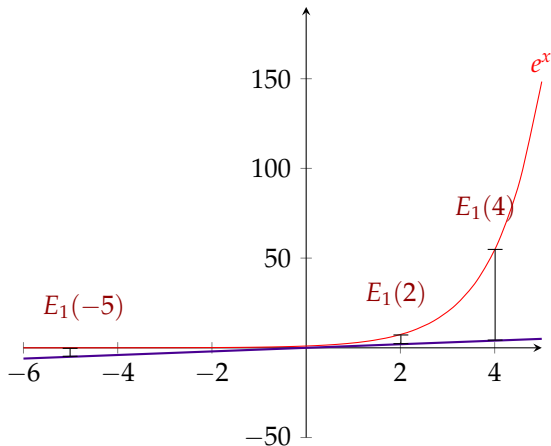
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- ▶ But, it's not immediately clear whether  $e^x \stackrel{?}{=} \sum_{n=0}^{\infty} \frac{x^n}{n!}$ .  
How could we determine this?
- ▶

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ \iff 0 &= e^x - \sum_{n=0}^{\infty} \frac{x^n}{n} = e^x - \lim_{n \rightarrow \infty} \underbrace{\sum_{k=0}^n \frac{x^k}{k!}}_{T_n(x)} = \lim_{n \rightarrow \infty} \underbrace{[e^x - T_n(x)]}_{E_n(x)} \\ \iff 0 &= \lim_{n \rightarrow \infty} E_n(x) \quad (\text{for all } x) \end{aligned}$$

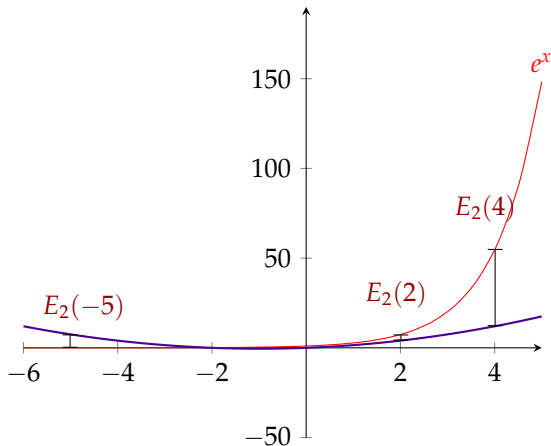
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If  $\lim_{n \rightarrow \infty} E_n(x) = 0$  for all  $x$ , then  $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  for all  $x$ .



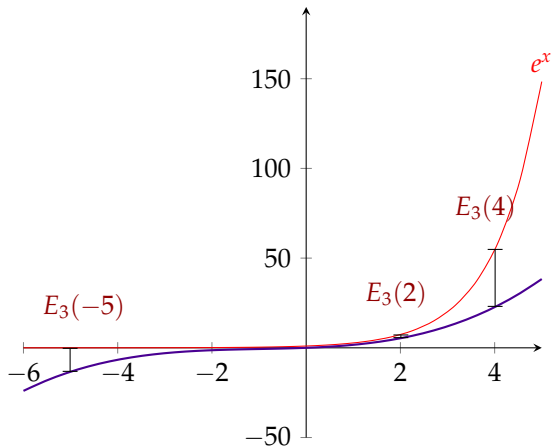
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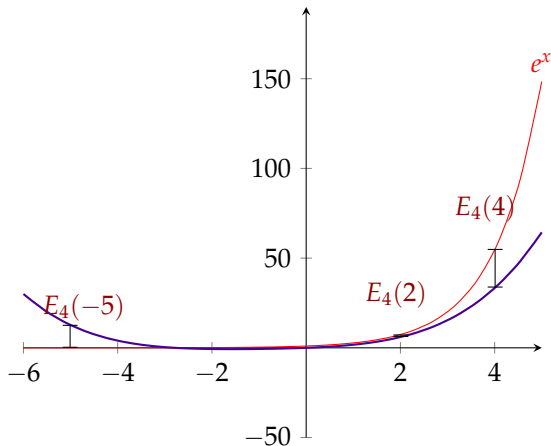
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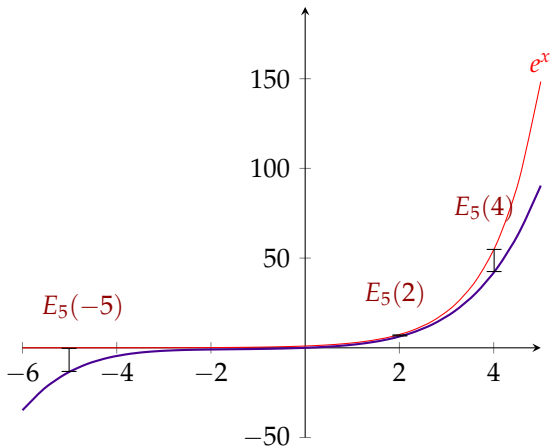
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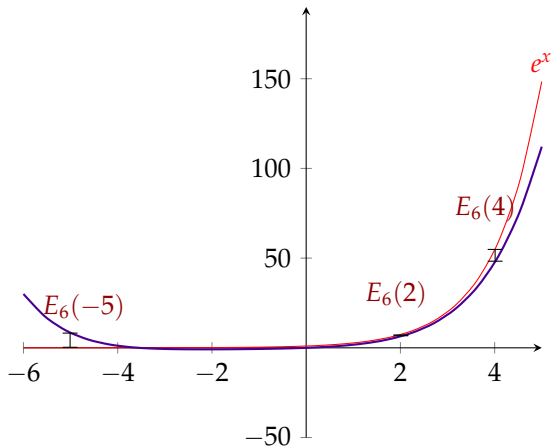
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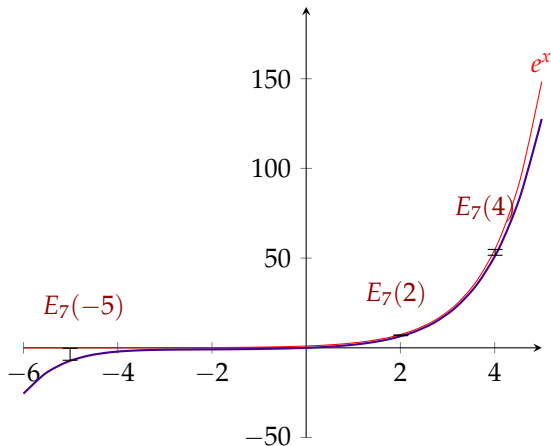
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If  $\lim_{n \rightarrow \infty} E_n(x) = 0$  for all  $x$ , then  $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$  for all  $x$ .

It *looks* plausible, especially when  $x$  is close to 0. Let's try to prove it.



## Equation 3.6.1-b

Let  $T_n(x)$  be the  $n$ -th order Taylor approximation of a function  $f(x)$ , centred at  $a$ . Then  $E_n(x) = f(x) - T_n(x)$  is the error in the  $n$ -th order Taylor approximation.

For some  $c$  strictly between  $x$  and  $a$ ,

$$E_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c) \cdot (x-a)^{n+1}$$

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When  $f(x) = e^x$ ,

$$E_n(x) = e^c \frac{x^{n+1}}{(n+1)!}$$

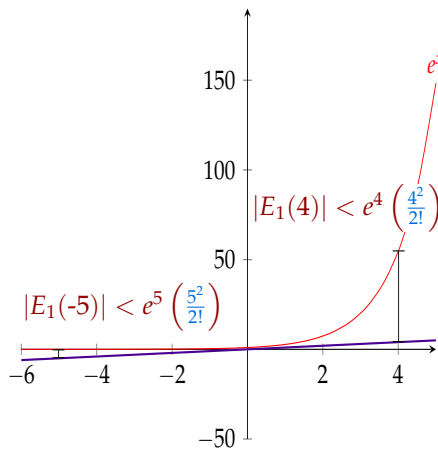
for some  $c$  between 0 and  $x$ .

$$E_n(x) = e^x - T_n(x)$$

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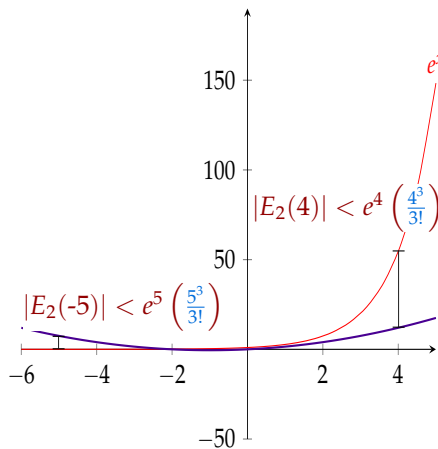
We found  $0 \leq |E_n(x)| < e^{|x|} \frac{|x|^{n+1}}{(n+1)!}$  for large  $n$ , hence  $\lim_{n \rightarrow \infty} |E_n(x)| = 0$ .



For a particular value of  $x$ :

We saw  $0 = \lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!}$

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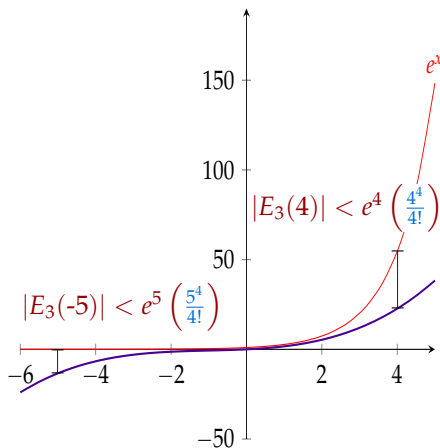
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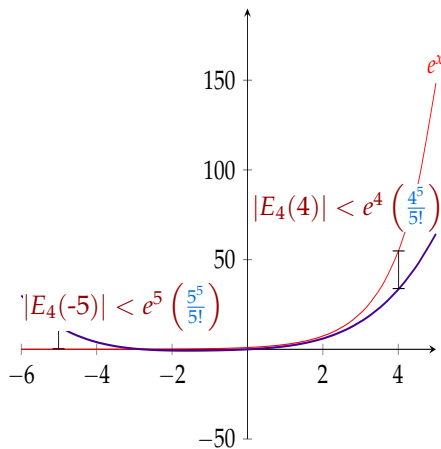
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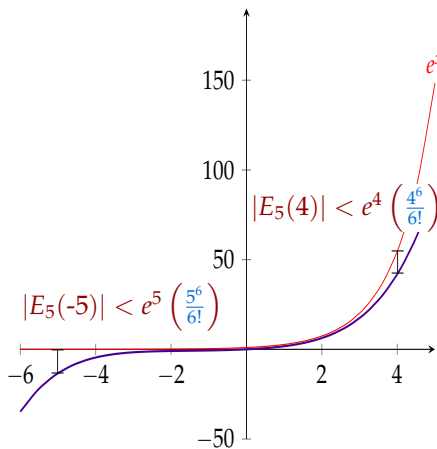
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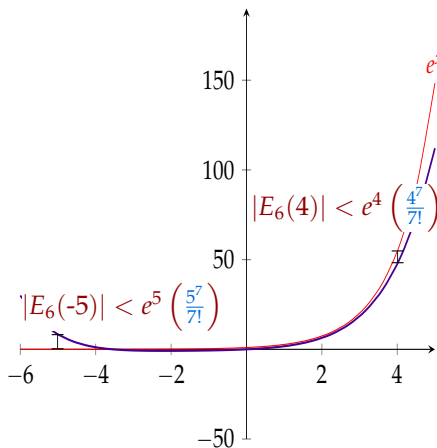
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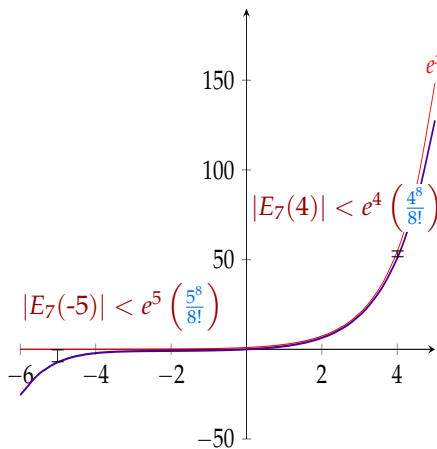
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# TAYLOR POLYNOMIAL ERROR FOR SINE AND COSINE

## Equation 3.6.1-b

Let  $T_n(x)$  be the  $n$ -th order Taylor approximation of a function  $f(x)$ , centred at  $a$ . Then  $E_n(x) = f(x) - T_n(x)$  is the error in the  $n$ -th order Taylor approximation.

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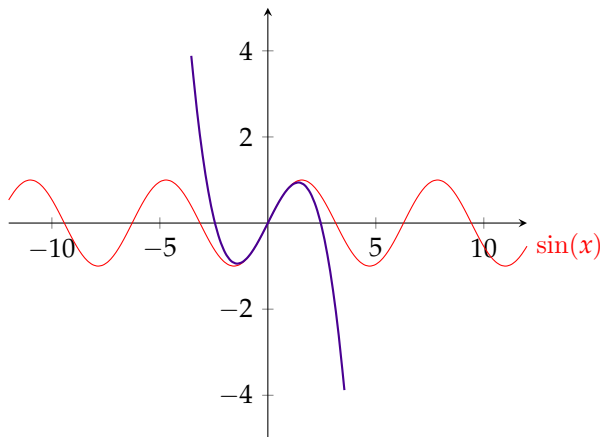
Suppose  $f(x)$  is either  $\sin x$  or  $\cos x$ . Is  $f(x)$  equal to its Maclaurin series?

# TAYLOR POLYNOMIAL ERROR FOR SINE AND COSINE

Suppose  $f(x)$  is either  $\sin x$  or  $\cos x$ .

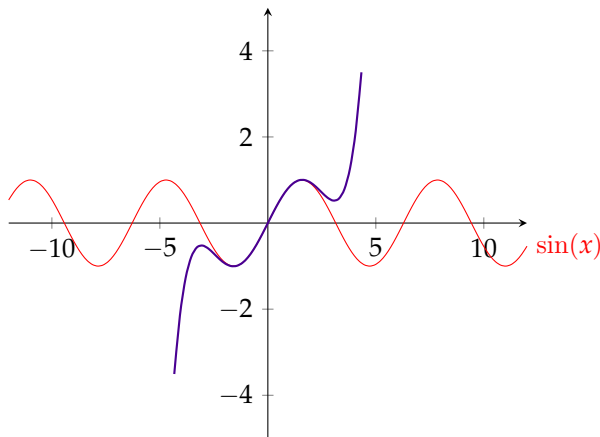
$$|E_n(x)| = \frac{1}{(n+1)!} \left| f^{(n+1)}(c) \right| |x|^{n+1}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



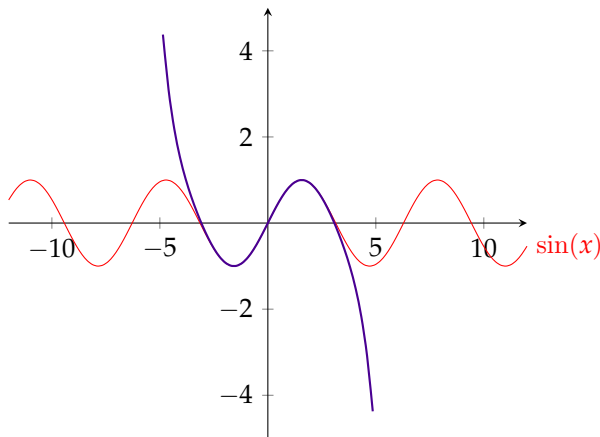
$$T_3(x) = x - \frac{x^3}{3!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



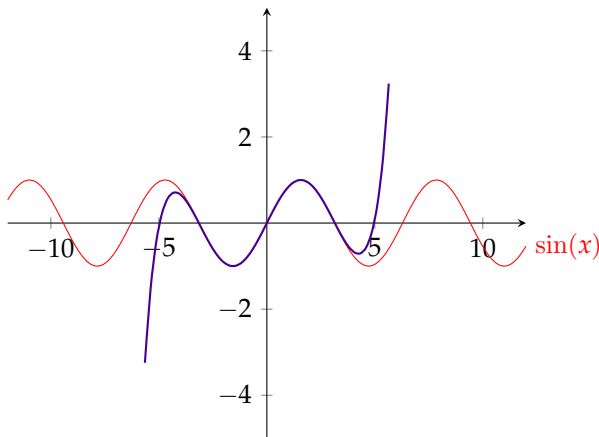
$$T_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



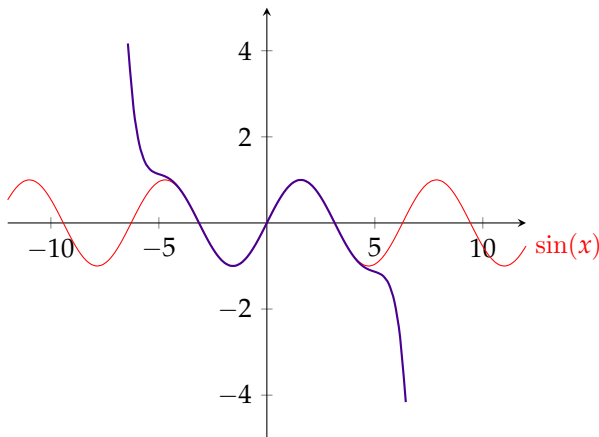
$$T_7(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



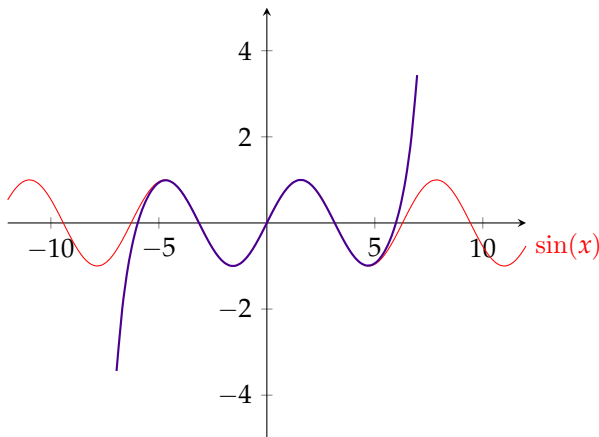
$$T_9(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



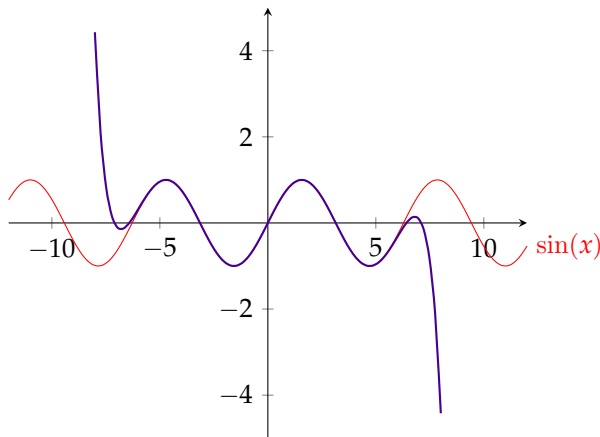
$$T_{11}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



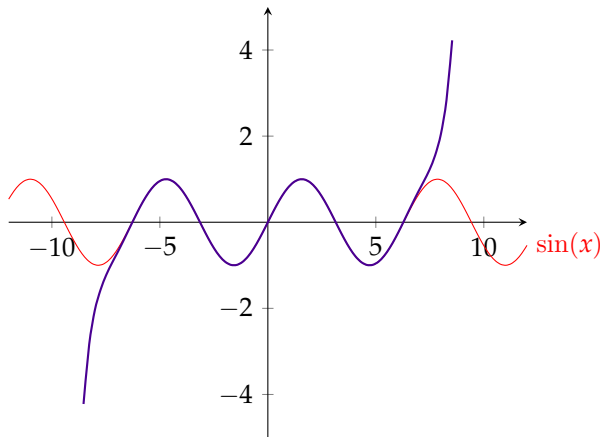
$$T_{13}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \frac{x^{13}}{13!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



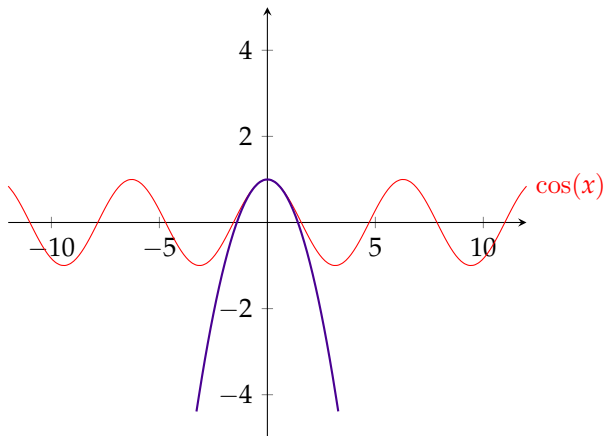
$$T_{15}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \frac{x^{13}}{13!} - \frac{x^{15}}{15!}$$

# TAYLOR POLYNOMIALS FOR $\sin(x)$



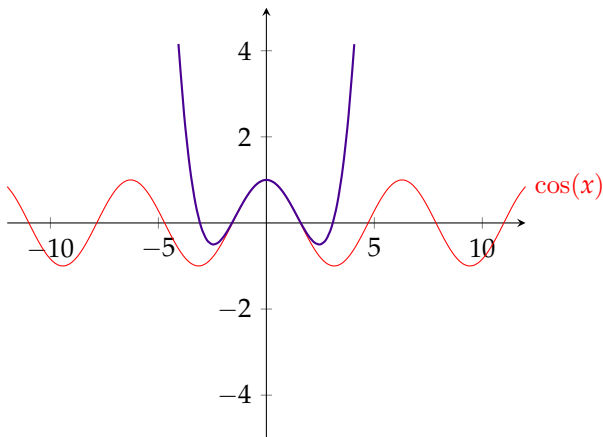
$$T_{17}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \frac{x^{13}}{13!} - \frac{x^{15}}{15!} + \frac{x^{17}}{17!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



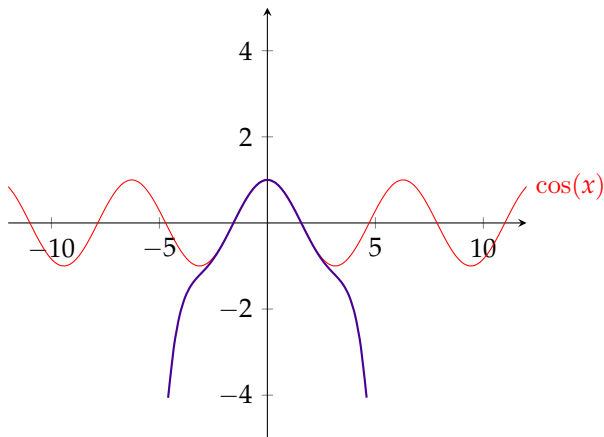
$$T_2(x) = 1 - x^2$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



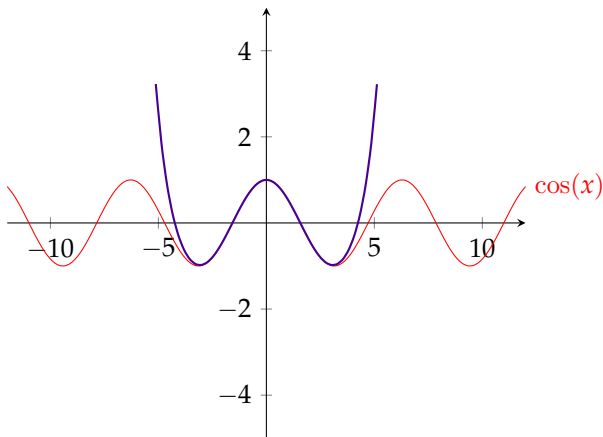
$$T_4(x) = 1 - x^2 + \frac{x^4}{4!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



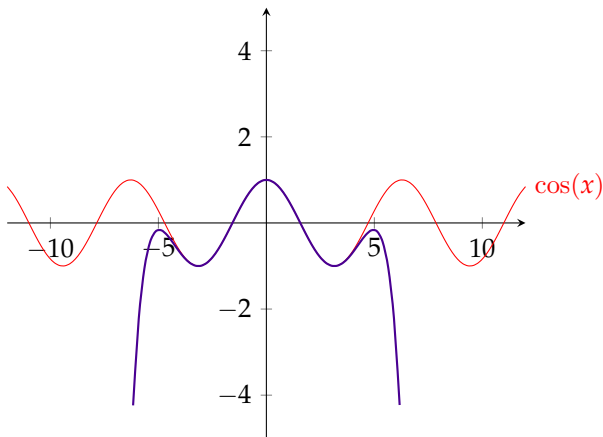
$$T_6(x) = 1 - x^2 + \frac{x^4}{4!} - \frac{x^6}{6!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



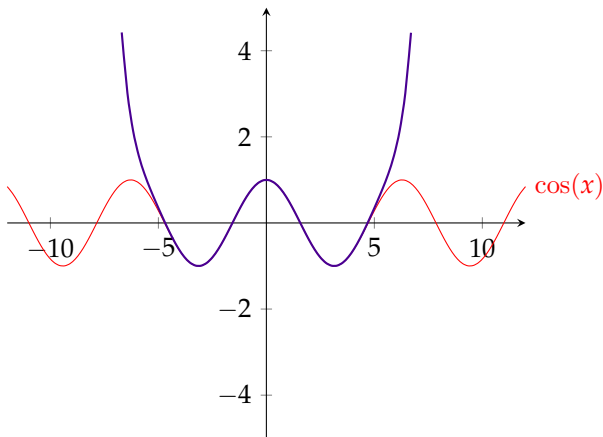
$$T_8(x) = 1 - x^2 + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



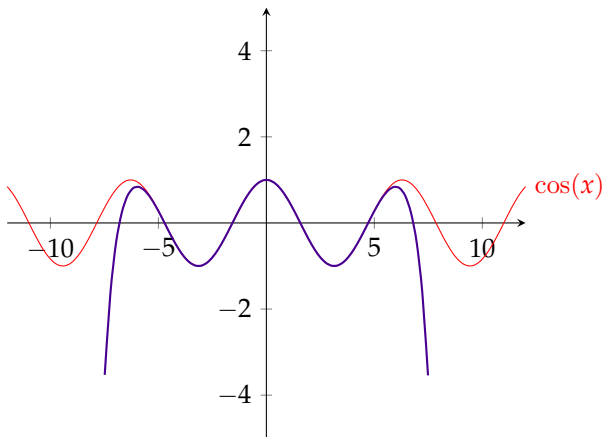
$$T_{10}(x) = 1 - x^2 + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



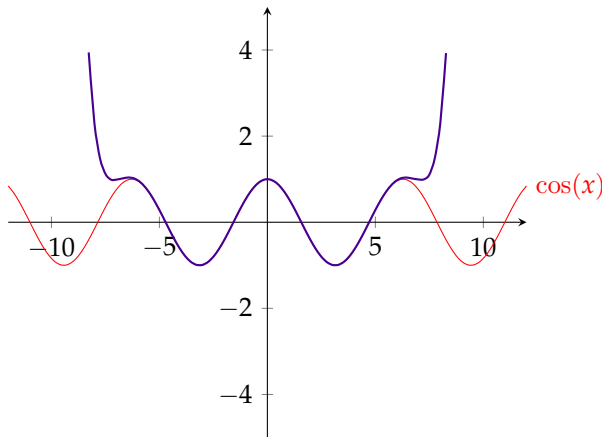
$$T_{12}(x) = 1 - x^2 + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \frac{x^{12}}{12!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



$$T_{14}(x) = 1 - x^2 + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \frac{x^{12}}{12!} - \frac{x^{14}}{14!}$$

# TAYLOR POLYNOMIALS FOR $\cos(x)$



$$T_{16}(x) = 1 - x^2 + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \frac{x^{12}}{12!} - \frac{x^{14}}{14!} + \frac{x^{16}}{16!}$$

## Selected Taylor series that equal their functions

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{for all } -\infty < x < \infty$$

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!} x^{2n+1} \quad \text{for all } -\infty < x < \infty$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n)!} x^{2n} \quad \text{for all } -\infty < x < \infty$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{for all } -1 < x < 1$$

$$\log(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} \quad \text{for all } -1 < x \leq 1$$

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad \text{for all } -1 \leq x \leq 1$$

# COMPUTING $\pi$

Use the fact that  $\arctan 1 = \frac{\pi}{4}$  to find a series converging to  $\pi$  whose terms are rational numbers.

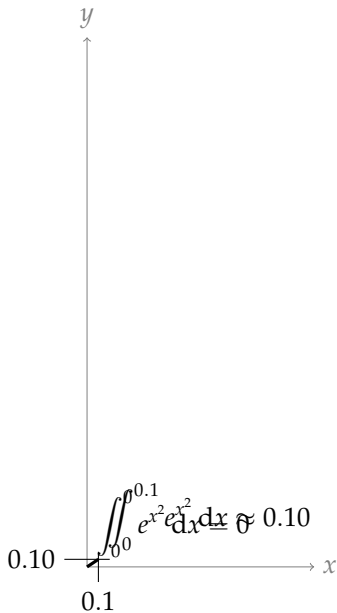
# ERROR FUNCTION

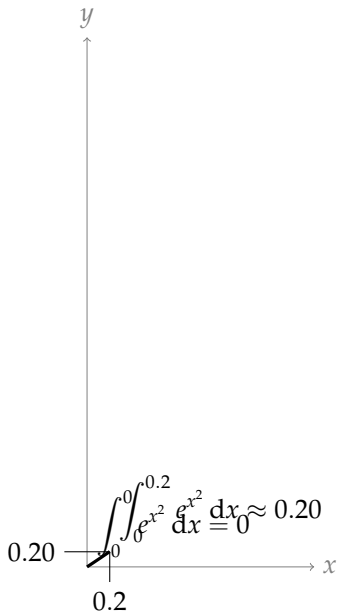
The *error function*

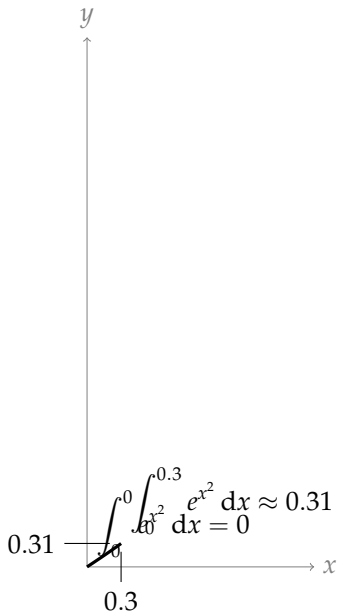
$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

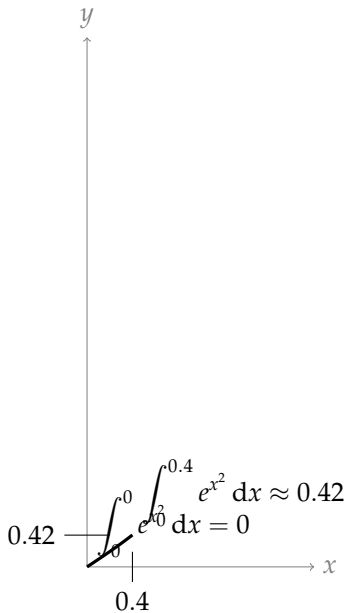
is used in computing “bell curve” probabilities.

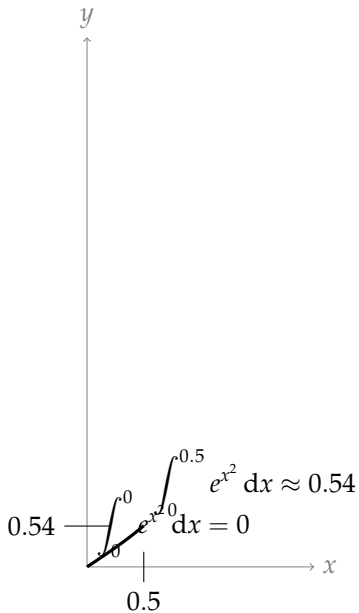


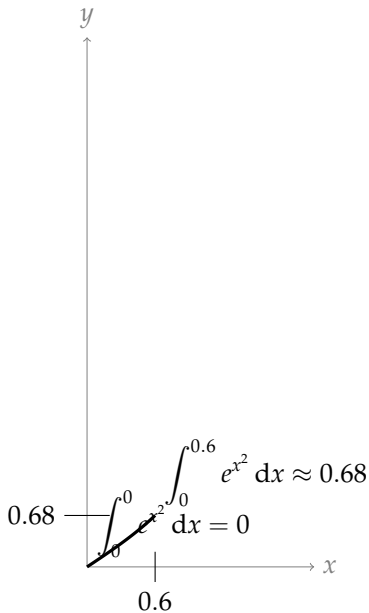


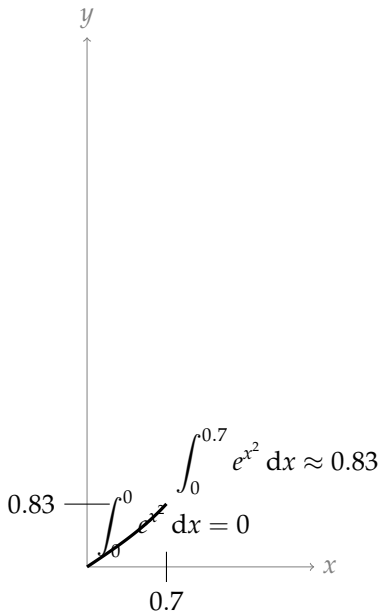


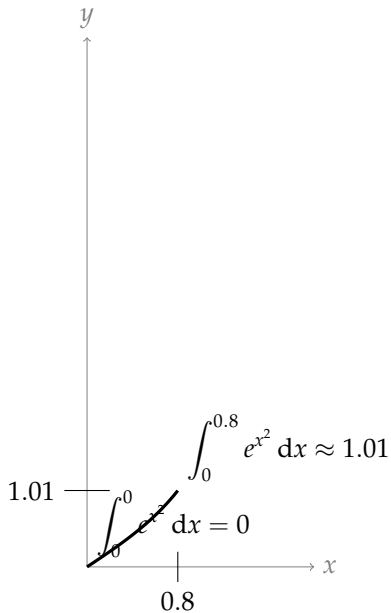
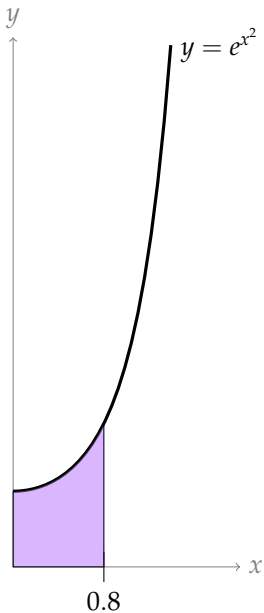


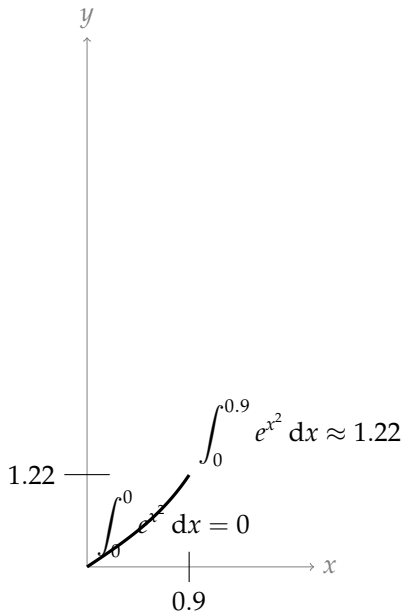
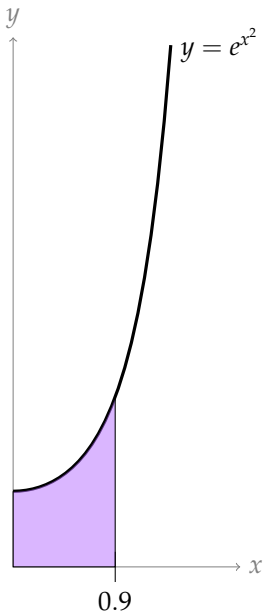


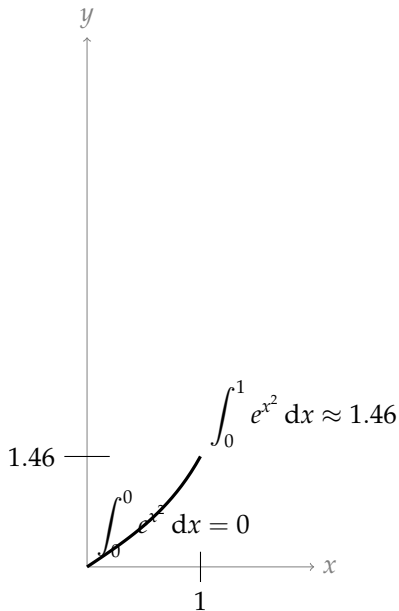
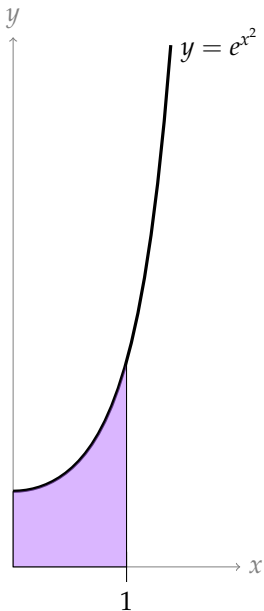


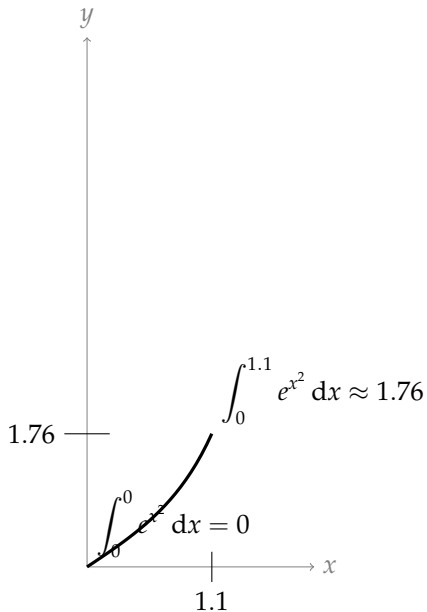
















# ERROR FUNCTION

The *error function*

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

is used in computing “bell curve” probabilities.

The indefinite integral of the integrand  $e^{-t^2}$  cannot be expressed in terms of standard functions. But we can still evaluate the integral to within any desired degree of accuracy by using the Taylor expansion of the exponential.

For example, evaluate  $\operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)$ .



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# EVALUATING A CONVERGENT SERIES

Evaluate  $\sum_{n=1}^{\infty} \frac{1}{n \cdot 3^n}$



# FINDING A HIGH-ORDER DERIVATIVE

Let  $f(x) = \sin(2x^3)$ . Find  $f^{(15)}(0)$ , the fifteenth derivative of  $f$  at  $x = 0$ .



Given that  $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ , we have a new way of evaluating the familiar limit

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} :$$

Evaluate  $\lim_{x \rightarrow 0} \frac{\arctan x - x}{\sin x - x}$ .

## Included Work

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